

Integrating The Integrated Circuit

Presenting the story of a chip that we cannot now live without, which is an interesting outcome of two engineers who developed it without knowing each other

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Unlike the invention of television which was replete with snooping, spying and courtroom drama, invention of integrated circuit (IC) was the outcome of two engineers who developed it separately without knowing each other and a host of unsung heroes.

IC is an invention that changed the way of the world forever. As usual, the Nobel committee took its time to award a Nobel and finally the award came in the year 2000 for an invention of 1958. The Nobel winner wrote in his autobiography submitted to the Nobel committee, "I would like to mention another right person at the right time, namely, Robert Noyce, a contemporary of mine who worked at Fairchild Semiconductor. While Robert and I followed our own paths, we worked hard together to achieve commercial acceptance for ICs. If he were still alive, I have no doubt we would have shared this prize." In this world torn with jealousy, personal egos, profits and politics, salute the unassuming inventor Jack Kilby!

And the tribute for him? Unprecedented growth of ICs. The very first IC contained only four transistors, and the present-day chip Core i5 contains 995 million transistors. He "didn't realise

then that the integrated circuit would reduce the cost of electronic functions by a factor of a million to one."

Transistor was an outstanding invention which revolutionised electronics. But building complex circuits required a large number of transistors and other passive components. Think of those long-standing ICs, 741 and 555. The ubiquitous 741 op-amp designed by Dave Fullagar in Fairchild Semiconductor in 1968. 555 IC was first introduced around 1971 as 'The IC Time Machine.' Designed in 1971 by Hans Camenzind under contract to Signetics, it still sells about one billion units every year. Think of wiring those 741 or 555 ICs individually which have 20 and 28 transistors, respectively. It is a mammoth task and sheer 'tyranny of numbers.'

It was in this tyrannical scenario that Jack Kilby joined the semiconductor lab at Texas Instruments in 1958. Soon he was asked to develop smaller electrical circuits, kind of micro-modules, specifically for the military. As he proceeded with his task, he was not convinced that the micro-module was the answer—still a large number of components needed to be hardwired.

The beginning

Three problems were bogging down the development of microelectronics:

integration, isolation and connection. There was no way of integrating all different active and passive components on a single semiconductor crystal. Even if connected, there was no way to electrically isolate them. Also, there was no way to connect individual components, at best they could be done with gold wires.

Geoffrey Dummer thought otherwise, "With the advent of the transistor and the work on semiconductors generally, it now seems possible to envisage electronic equipment in a solid block with no connecting wires. The block may comprise layers of insulating, conducting, rectifying and amplifying materials, the electronic functions being connected directly by cutting out areas of the various layers."

He said so in his paper at the US Electronic Components Symposium. Geoffrey William Arnold Dummer, a British electronics engineer and consultant, is credited as 'The Prophet of the Integrated Circuit.'

Kilby thought likewise; he summed up the thoughts in his mind of those days in a later day in the year 1976's article titled 'Invention of the IC,' thus, "Further thought led me to the conclusion that semiconductors were all that were really required—that resistors and capacitors (passive devices), in particular, could be made from the same material as the active devices (transistors). I also realised that, since all of the components could be made of a single material, they could also be made *in situ* interconnected to form a complete circuit." He began sketching his ideas.

Provisionally in July 1958, he was alone in the deserted laboratory as the



Intel Core i7 processor (Courtesy: www.intel.com)